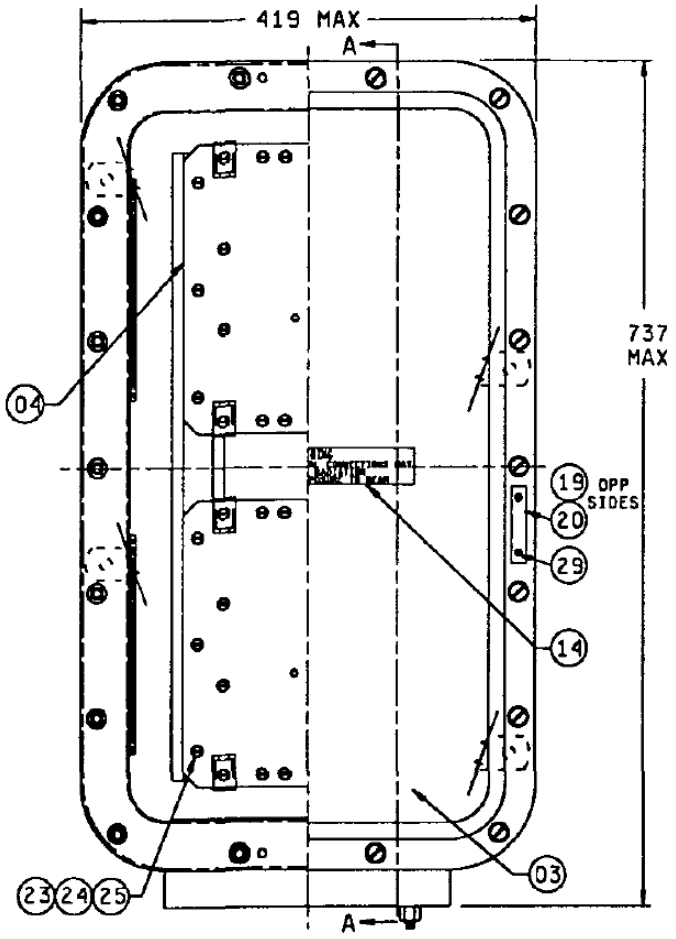
METRIC
MIL-DTL-24728/2C
w/AMENDMENT 1
24 March 2023
SUPERSEDING
MIL-DTL-24728/2C
18 September 2018

DETAIL SPECIFICATION SHEET

INTERCONNECTION BOX, FIBER OPTIC, SUBMERSIBLE, 308.4 X 609.6 MM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24728.



ABOVE COVER SHOWN PARTIALLY REMOVED

FIGURE 1. Enclosure assembly, M24728/2-005 and M24728/2-006.

AMSC N/A FSC 6099

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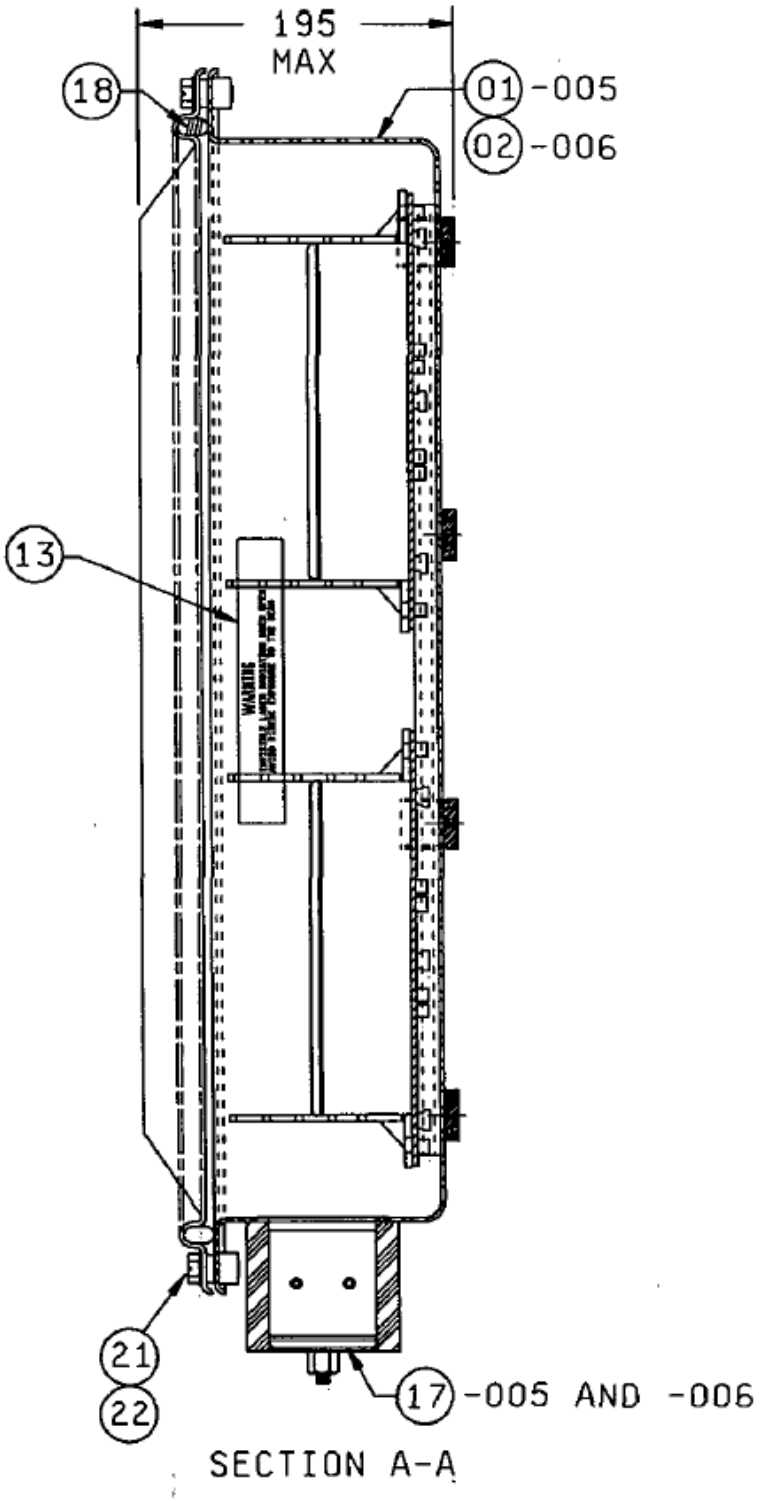
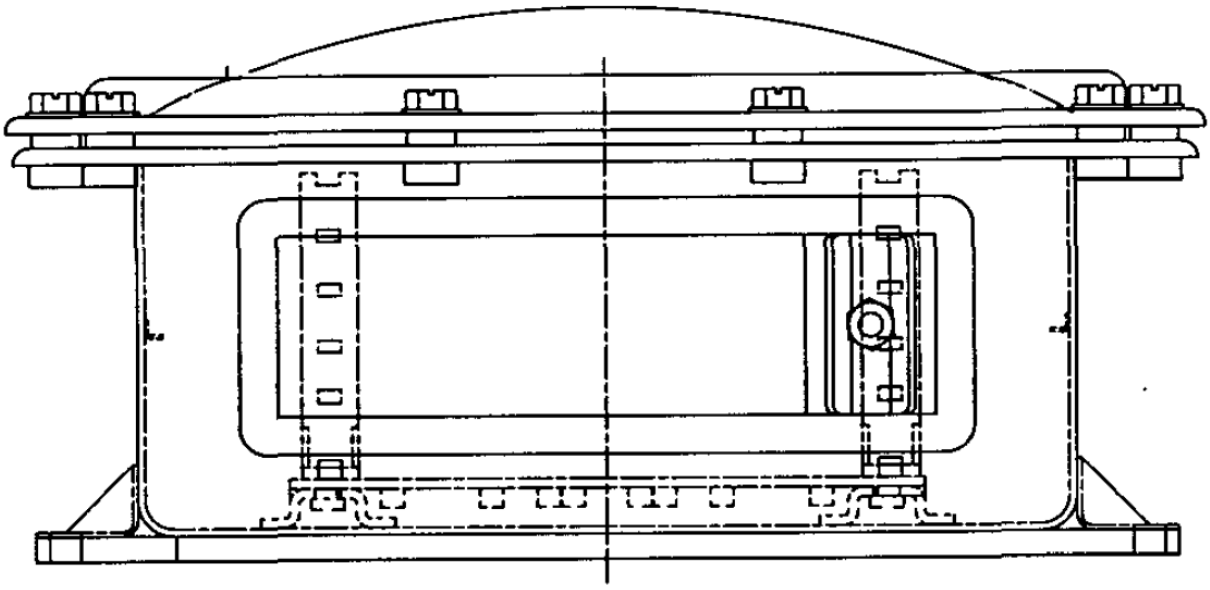


FIGURE 1. Enclosure assembly, M24728/2-005 and M24728/2-006 – Continued.

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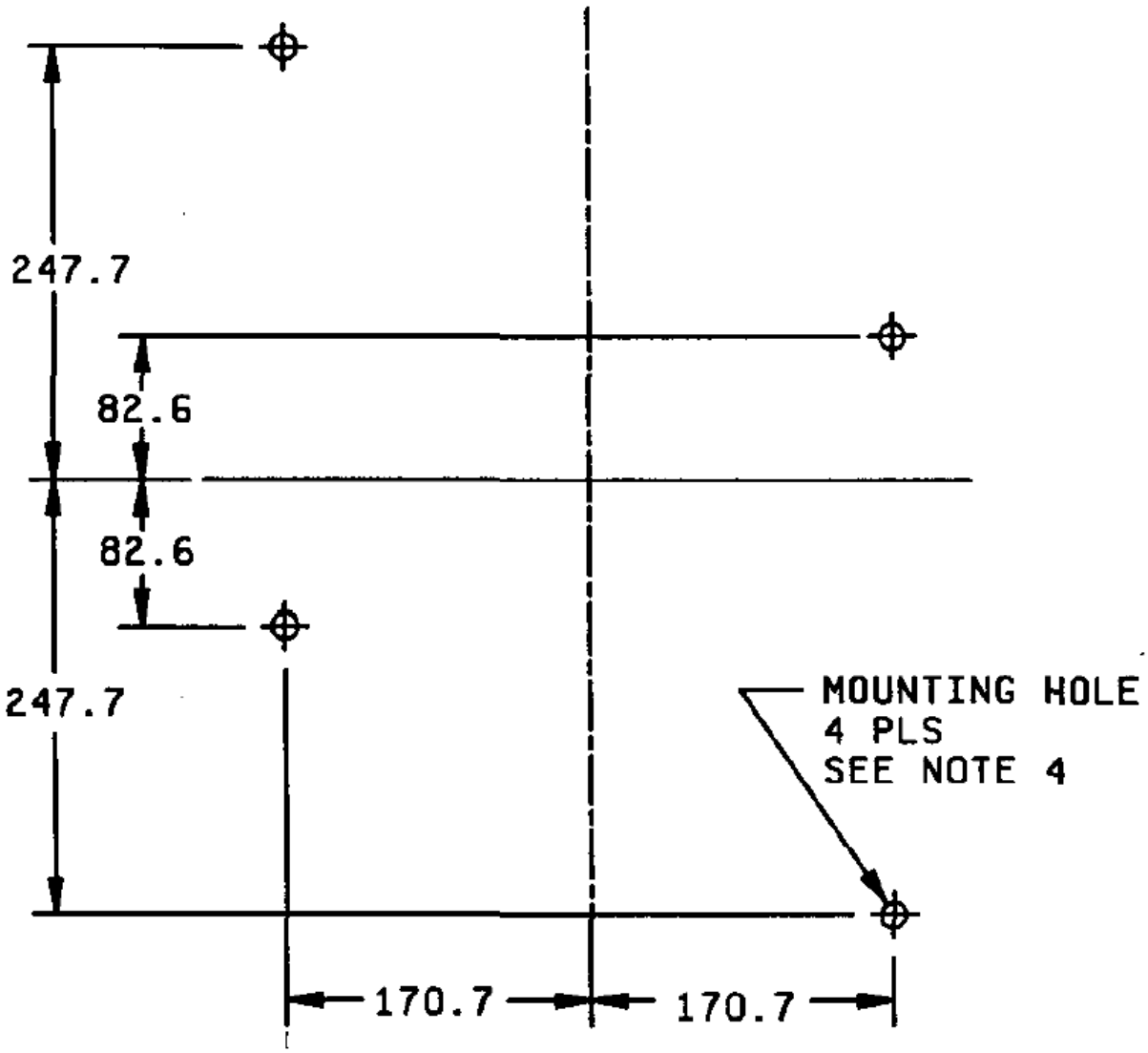
Millimeters	Inches
195	7.7
419	16.5
737	29.0

NOTES:

1. Dimensions are in millimeters.
2. Inch-pound equivalents are given for general information only.
3. DELETED.
4. For identification of subcomponent parts (circled numbers), see table I.

FIGURE 1. Enclosure assembly, M24728/2-005 and M24728/2-006 – Continued.

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Millimeters	Inches
82.6	3.25
170.7	6.72
247.7	9.75

NOTES:

1. Dimensions are in millimeters.
2. Inch-pound equivalents are given for general information only.
3. Dimensions are controlled by welding or brazing.
4. Mounting holes shall accommodate M12.7 x 1.95 (0.50-13) bolts.

FIGURE 2. Box mounting hole dimensions.

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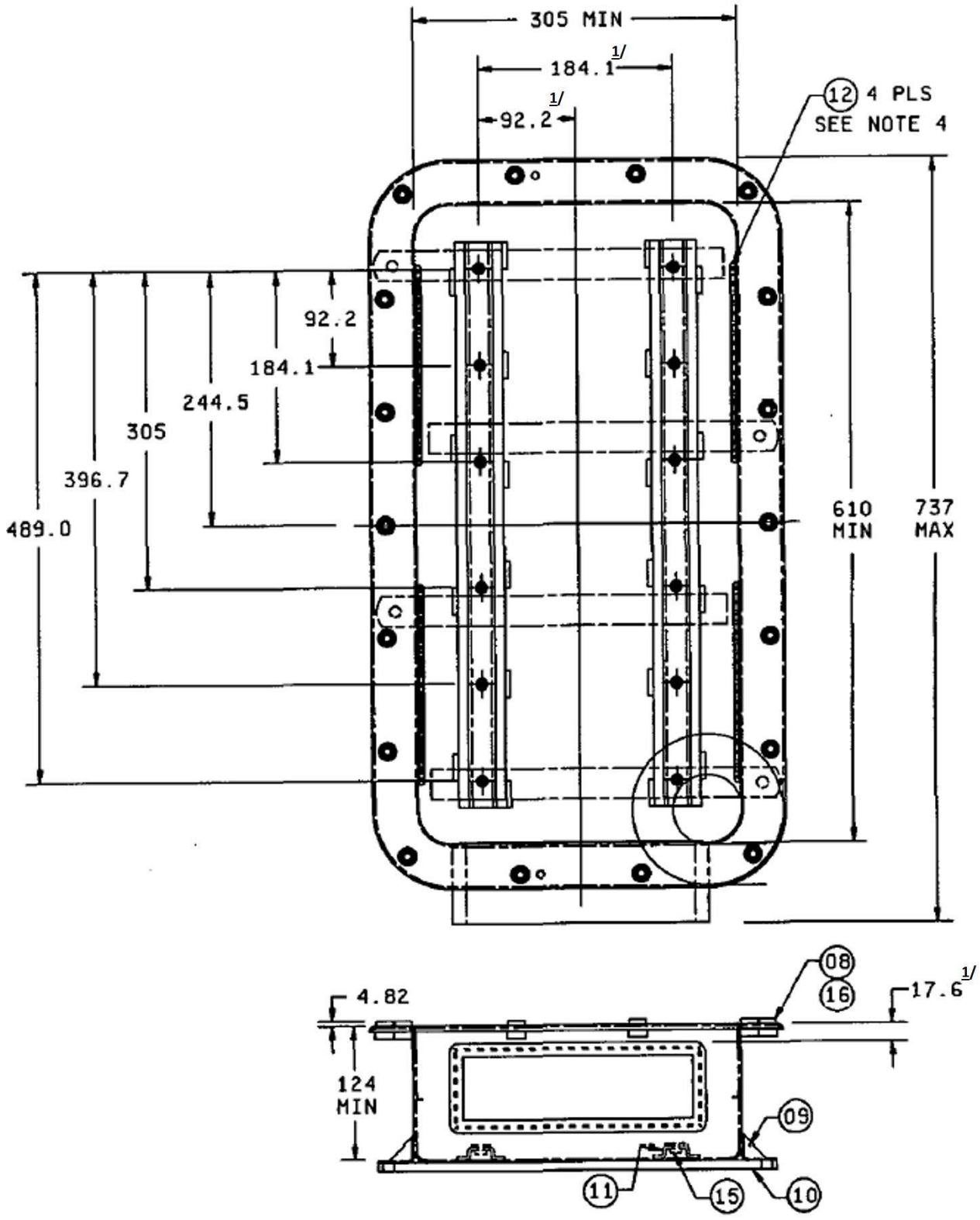
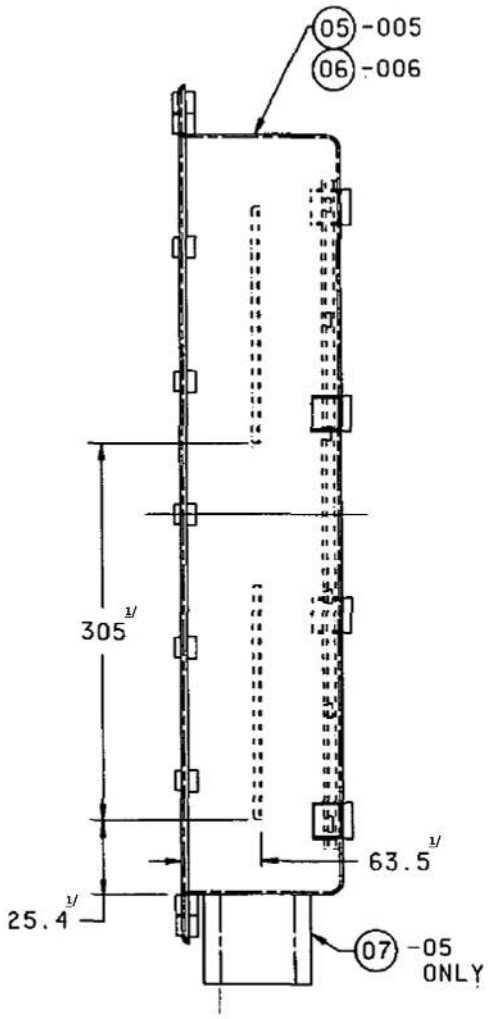


FIGURE 3. Box assembly.

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Millimeters	Inches	Millimeters	Inches
4.82	0.190	184.1	7.25
17.6	0.69	305	12.01
25.4	1.00	396.7	15.62
63.5	2.50	489	19.25
92.2	3.63	610	24.0
124.0	4.9	737	29.0

FOOTNOTE:

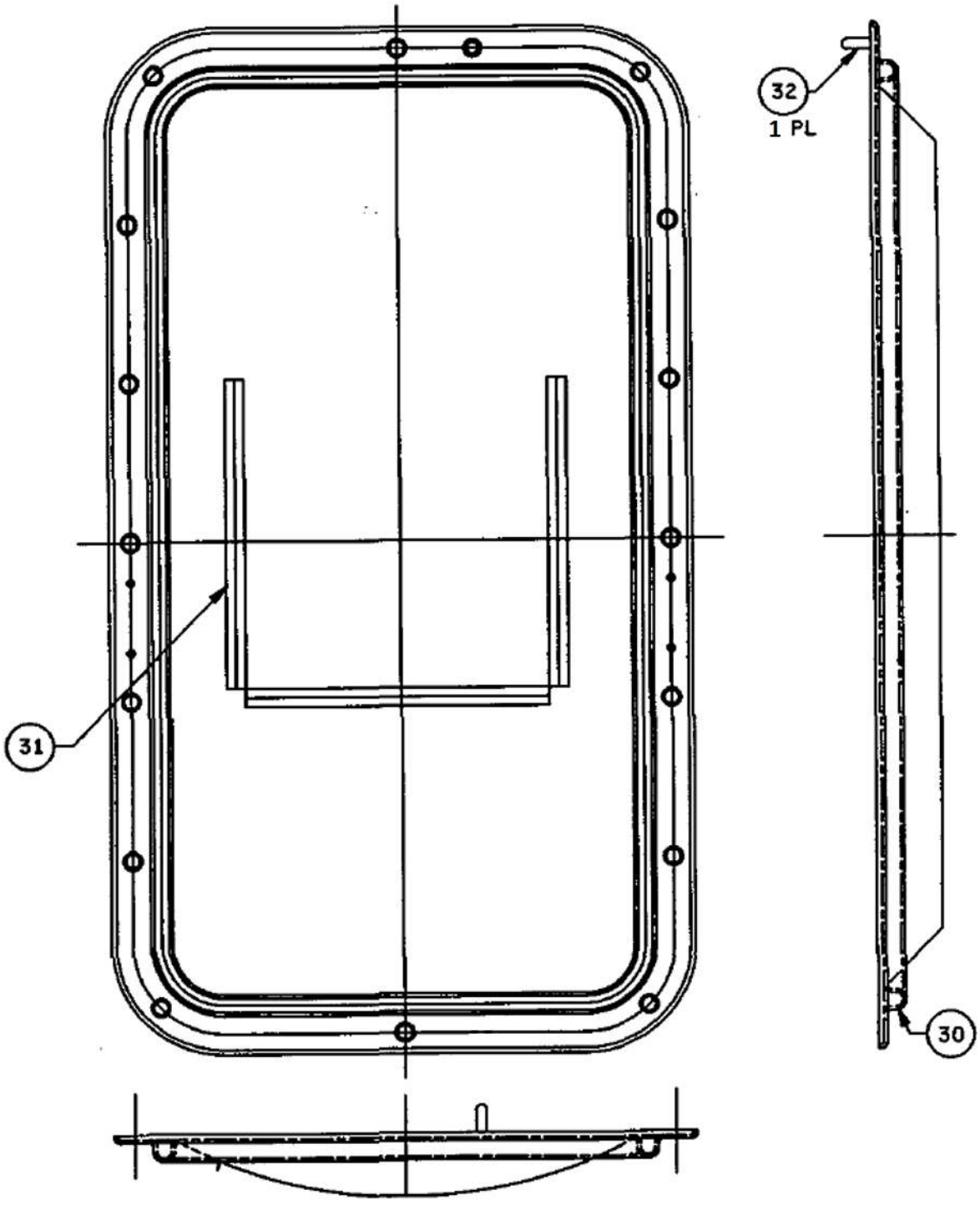
1/ Dimensions are controlled by welding or brazing.

NOTES:

1. Dimensions are in millimeters.
2. Inch-pound equivalents are given for general information only.
3. DELETED.
4. For identification of subcomponent parts (circled numbers), see table I.

FIGURE 3. Box assembly – Continued.

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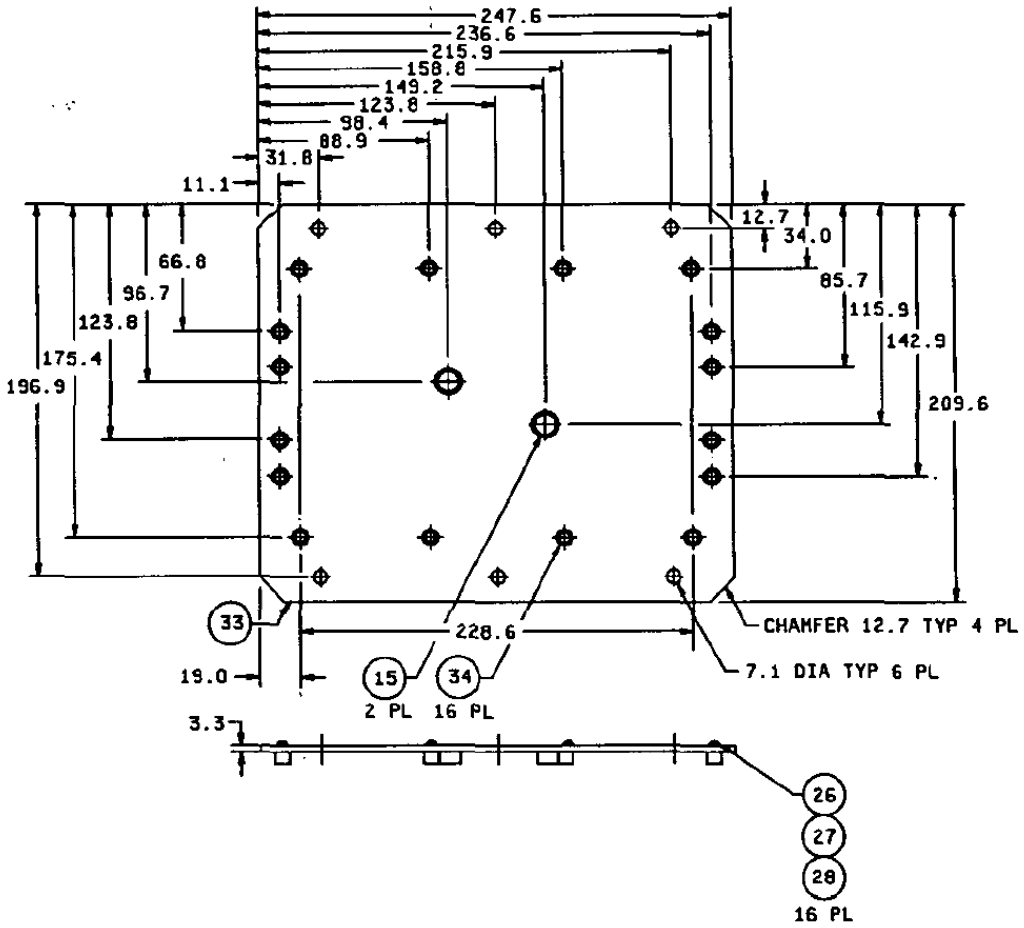


NOTES:

1. Detailed layout and placement of diagram holder is not specified.
2. For identification of subcomponent parts (circled numbers), see table I.

FIGURE 4. Cover assembly.

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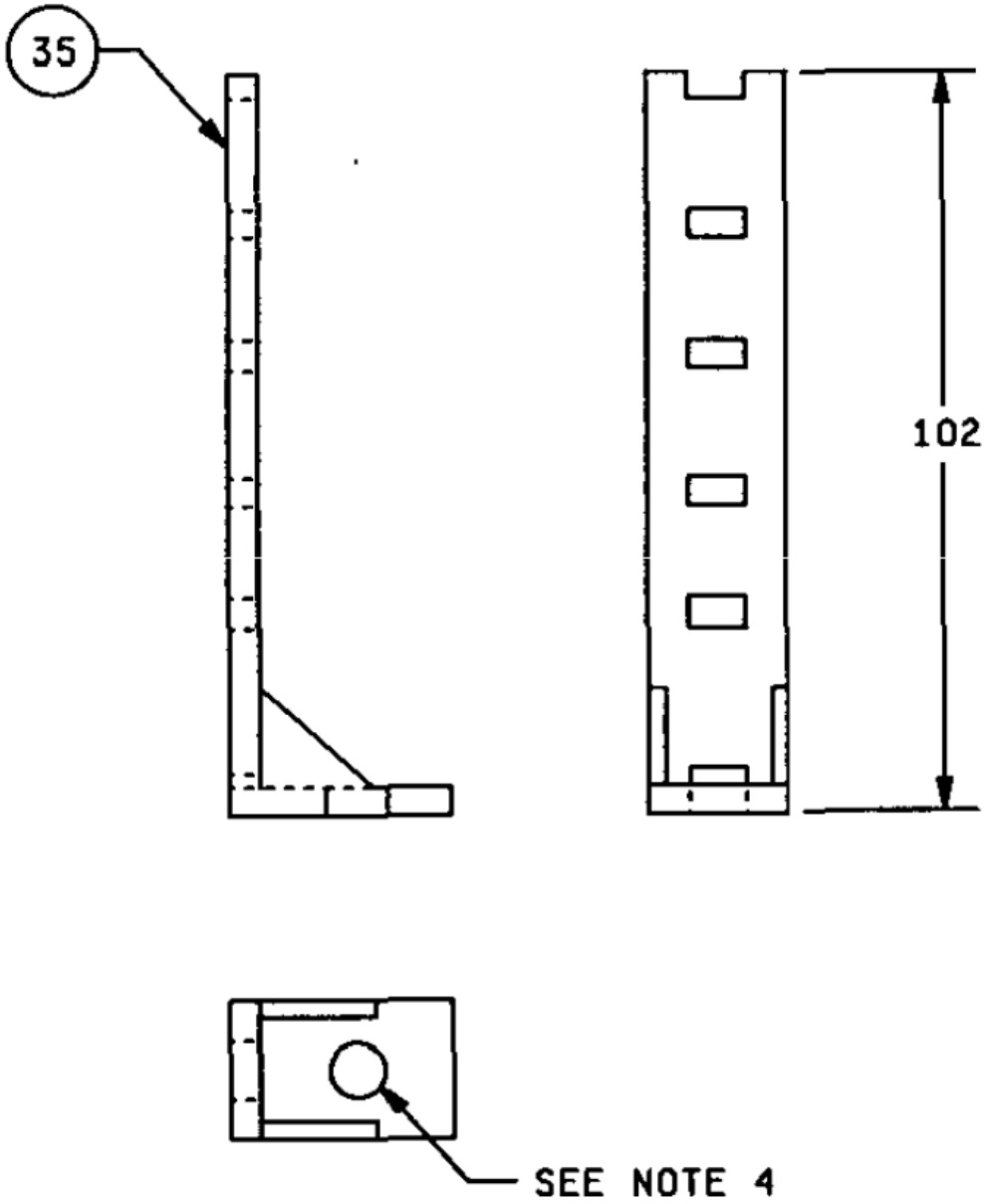
Millimeters	Inches	Millimeters	Inches	Millimeters	Inches
3.3	0.13	88.9	3.50	175.4	6.91
7.1	0.28	96.7	3.69	196.9	7.75
11.2	0.44	98.4	3.88	209.6	8.25
12.7	0.50	115.9	4.56	215.9	8.50
19.0	0.75	123.8	4.88	228.6	9.00
31.8	1.25	142.9	5.63	236.6	9.31
34.0	1.34	149.2	5.88	247.6	9.75
64.1	2.53	158.8	6.25		
85.7	3.38				

NOTES:

1. Dimensions are in millimeters.
2. Inch-pound equivalents are given for general information only.
3. DELETED.
4. For identification of subcomponent parts (circled numbers), see table I.

FIGURE 5. Mounting plate assembly.

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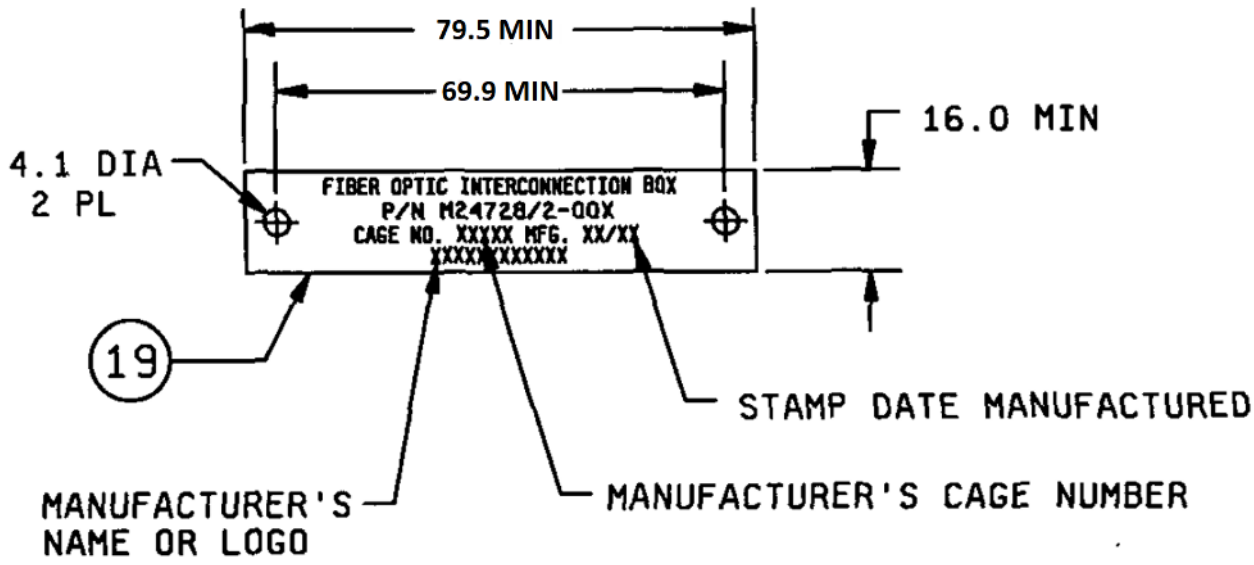
Millimeters	Inches
102	4.0

NOTES:

1. Dimensions are in millimeters.
2. Inch-pound equivalents are given for general information only.
3. DELETED.
4. Hole is 5.6 mm (0.22 inch) diameter.
5. For identification of subcomponent parts (circled numbers), see table I.

FIGURE 6. Tie wrap post.

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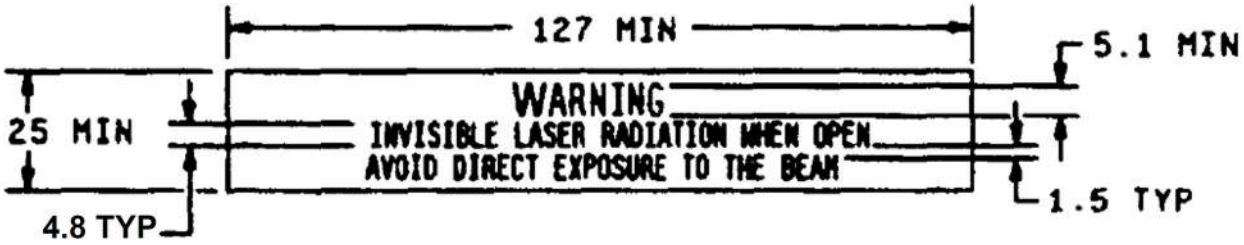
Millimeters	Inches
4.1	0.16
16.0	0.63
69.9	2.75
79.5	3.13

NOTES:

1. Dimensions are in millimeters.
2. Inch-pound equivalents are given for general information only.
3. DELETED.
4. For identification of subcomponent parts (circled numbers), see table I.

FIGURE 7. Identification plate.

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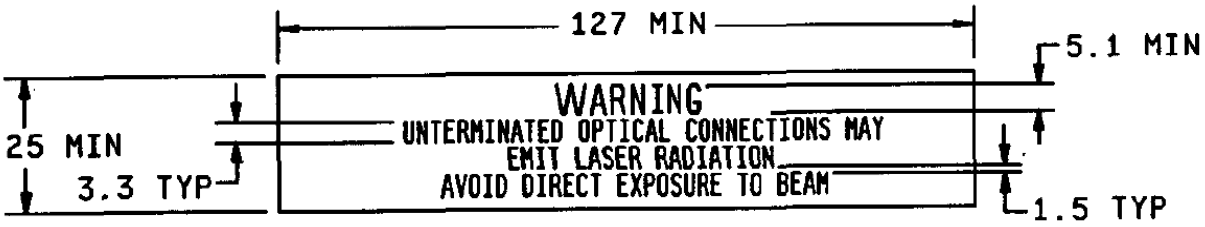
Millimeters	Inches
1.5	0.06
4.8	0.19
5.1	0.20
25	1.0
127	5.0

NOTES:

1. Dimensions are in millimeters.
2. Inch-pound equivalents are given for general information only.
3. The signal word “WARNING” shall be in yellow on a black background in the upper panel. The lower panel shall consist of black letters on a yellow background.
4. DELETED.

FIGURE 8. Inside warning label.

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Millimeters	Inches
1.5	0.06
3.3	0.13
5.1	0.20
25	1.0
127	5.0

NOTES:

1. Dimensions are in millimeters.
2. Inch-pound equivalents are given for general information only.
3. The signal word “WARNING” shall be in yellow on a black background in the upper panel. The lower panel shall consist of black letters on a yellow background.
4. DELETED.

FIGURE 9. Outside warning label.

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TABLE I. Parts list.

Subcomponent identification numbers, circled (see figures 1 through 9)	Part name	PIN M24728/2 dash number		Subassembly number				Remarks and reference figures
		-005	-006	01	02	03	04	
01	Box assembly with MCT frame	1		1				1 and 3
02	Box assembly without MCT frame		1		1			1 and 3
03	Cover assembly	1	1			1		1 and 4
04	Mounting plate assembly	2	2				1	5
05 1/, 2/	308.4 x 609.6 Box with MCT frame			1				3
06 1/, 2/	308.4 x 609.6 Box without MCT frame				1			3
07 3/, 4/	MCT frame			1				3
08 3/, 2/	Flange pad			18	18			3
09 2/, 5/	Gusset			4	4			3
10 3/	Mounting leg			4	4			3
11 3/	Mounting channel			2	2			3
12 3/	Tie wrap bracket			4	4			3
13 6/	Inside warning label	1	1					1 and 8
14 6/	Outside warning label	1	1					11 and 9
15 7/	M6.35 x 1.27 (0.25-20) insert			12	12		2	3 and 5
16 7/	M9.53 x 1.59 (0.375-16) helicoil			18	18			3
17 4/	MCT wedge pack	1	1					1
18 2/	Gasket	1	1					1
19 8/	Identification plate	1	1					1 and 7
20 9/	Blank nameplate	1	1					1
21 2/, 7/	M9.53 x 1.59 (0.375-16) hex bolt	18	18					1
22 7/	9.53 (0.375) flat washer	18	18					1
23 7/	M6.35 x 1.27 (0.25-20) x 15.9 pan head machine screws (PHMS)	12	12					1
24 7/	6.35 (0.25) flat washer	12	12					1
25 7/	6.35 (0.25) split lock washer	12	12					1
26 7/	M4.0 x 0.79 (8-32) x 12.7 PHMS						12	5
27 7/	4.0 (#8) flat washer						12	5
28 7/	4.0 (#8) split lock washer						12	5

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TABLE I. Parts list – Continued.

Subcomponent identification numbers, circled. (see figures 1 through 9)	Part name	PIN M24728/2 dash number		Subassembly number				Remarks and reference figures
		-005	-006	01	02	03	04	
29 7/	M3.5 x 0.79 (6-32) x 6.35 PHMS	4	4					1
30 1/, 2/	308.4 x 609.6 cover	1	1					4
31 2/, 8/, 9/	Diagram holder					1		4
32 2/, 9/	Dowel pin					1		4
33 9/	Mounting plate						1	5
34 7/	M4.0 x 0.79 (8-32) insert						16	5
35	Tie wrap post	8	8					5

NOTES:

1/ Aluminum 3003 or 6061 in accordance with ASTM B209/B209M.
2/ See MIL-E-24142/6 for guidance.
3/ Aluminum 5052 H-32 or 6061 in accordance with ASTM B209/B209M.
4/ See MIL-DTL-24705/1 for guidance.
5/ Aluminum 5052 H-32 or 6061 in accordance with ASTM B209/B209M, 2.39 mm (0.09 inch) thick.
6/ In accordance with MIL-DTL-15024, type G, adhesive-backed, metal foil.
7/ Corrosion-resistant steel (CRES).
8/ Constructed to hold a 216- by 279-mm (8.5- by 11.0-inch) diagram.
9/ Aluminum 5052 H-32 in accordance with ASTM B209/B209M.

REQUIREMENTS:

Design and construction. See figures 1 through 9, and table I.

Interconnection box. The interconnection box shall be constructed in accordance with M24142/6-010 of MIL-E-24142/6.

Degree of enclosure. Submersible.

MCT. The Multiple Cable Transit (MCT) shall be constructed using MIL-DTL-24705/1 as guidance. The MCT interior dimensions shall be 217.4 by 60.0 millimeters (8.56 by 2.36 inches).

Enclosure assembly construction. The enclosure assembly shall be constructed using the following combinations of materials:

a. The box, MCT frame, flange pads, gussets, mounting legs, mounting channels, tie wrap bracket, and cover shall be aluminum 6061 in accordance with ASTM B209/B209M.

b. The box and cover shall be aluminum 3003 in accordance with ASTM B209/B209M. The MCT frame, flange pads, gussets, mounting legs, mounting channels, and tie wrap bracket shall be aluminum 5052 H-32 in accordance with ASTM B209/B209M.

Enclosure assemblies constructed using aluminum 6061 shall not be welded.

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Enclosure assemblies constructed using aluminum 6061 shall be dip-brazed in accordance with AWS C3.7M/C3.7.

Enclosure assemblies constructed using aluminum 6061 shall be artificially aged at 177 °C for 12 hours to a 6061-T4 condition.

Enclosure assemblies shall be class 1A conversion coated in accordance with MIL-DTL-5541 before being epoxy powder coated.

Cable interconnection interface. The cable feed-through shall be in accordance with MIL-DTL-24705 and shall accept cables having outer diameters between 6.1 and 25.4 millimeters (0.24 and 1.00 inch) constructed in accordance with MIL-PRF-85045.

Enclosure assembly stress relief. Enclosure assemblies shall be stress relieved in accordance with MIL-E-24142.

Mass: M24728/2-005: 11.3 kilograms (maximum).
M24728/2-006: 9.1 kilograms (maximum).

Interconnection box mounting. See figure 2.

Interconnect organization.

Fiber optic diagram holders. See figure 4.

Fiber optic cable tie posts. Eight (see figure 6).

Fiber optic splice tray holder. Two maximum (see MIL-DTL-24728/8).

Fiber optic connector module. Two maximum (see MIL-DTL-24728/6).

Fiber optic splices. When outfitted with 2 MIL-DTL-24728/8 splice tray holders, the interconnection box shall organize up to 48 MIL-PRF-24623/6 fiber optic splices per splice module (maximum of two 2 splice modules).

Fiber optic connector adapter. When outfitted with MIL-DTL-24728/6 connector modules, the interconnection box shall organize up to 48 MIL-DTL-83522/17 connector adapters (maximum of two connector modules).

Cable retention. Not applicable.

Cable seal flexing. Not applicable.

Cable twist. Not applicable.

Compression resistance. Not applicable.

Impact. Not applicable.

Water pressure. The applied pressure shall be 44.8 kilopascals (6.5 pounds per square inch) (equivalent to a depth of 15 feet). The interconnection cover bolts and the MCT wedge pack nut shall be tightened to a minimum torque of 16.94 newton-meters (150 inch-pounds) during this test.

Flammability. Not applicable.

Salt spray. Not applicable.

Flame spread. Not applicable.

Fungus test. The fungus test shall be conducted only on polymeric materials.

Operating and storage temperature range. See temperature range 1 of MIL-DTL-24728.

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Part or identifying number (PIN). See MIL-DTL-24728 and table I.

M24728/2-005, box, mounting plates, and lid, with cable entrance and wedge pack.

M24728/2-006, box, mounting plates, and lid, without cable entrance.

M24728/2-001, M24728/2-002, M24728/2-003, and M24728/2-004 have been cancelled without replacement.

Gasket size. The gasket size shall be 314.3 millimeters (12.375 inches) wide by 619.12 millimeters (24.375 inches) long. This corresponds to PIN M24142/6-010 of MIL-E-24142/6.

Insert blocks. Insert blocks compatible with the MCT frame shall be as specified in MIL-DTL-24705/1.

Measurement system. The U.S. Government preferred system of measurement is the metric (SI) system. However, since this item was originally designed using inch-pound units of measurement, in the event of conflict between the metric and inch-pound units, the inch-pound units shall take precedence.

AMENDMENT NOTATIONS: The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – SH
Air Force – 85
DLA – CC

Review activities:
Army – MI
Navy – AS, EC, YD
Air Force – 19
DLA – GS

Civil agency:
NASA – NA

Preparing activity:
Navy – SH
(Project 6099-2022-006)

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